

Deep Learning on sPHENIX Calorimeter Waveform Fits

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Introduction

i. The sPHENIX Detector

The sPHENIX detector is a cutting-edge jet detector, succeeding the PHENIX detector at RHIC in IP8. The main physics goal of sPHENIX is to further study the properties of the quark-gluon plasma (QGP) [1]. By measuring jets produced by different quarks, the effect of color charge and mass on the energy loss of quarks in the QGP can be inferred. To satisfy its physics goals, sPHENIX will take data for two years: p+p in 2024 and a large Au+Au run in 2025. sPHENIX is designed to take unprecedented amounts of minimum bias data at a high rate of 15kHz. This will result in a 200PB dataset for the 2025 Au+Au run containing about 100B events [2]. The processing of this data poses a unique challenge, so new avenues to efficiently reconstruct the data need to be investigated. Machine learning provides a novel approach to solving common computing problems faster than current techniques.

ii. sPHENIX Calorimetry

sPHENIX is composed of 10 sub-detectors (Figure 1), which include both calorimetry and tracking capabilities. sPHENIX calorimetry includes one electromagnetic calorimeter (EMCAL) and two hadronic calorimeters (iHCAL and oHCAL) [1]. The EMCAL detects particles that interact via electromagnetic interactions (photons and electrons/positrons), and the hadronic calorimeters

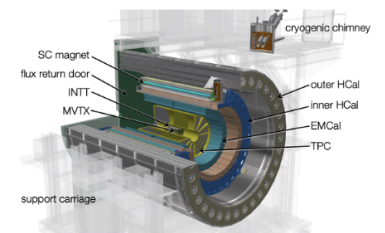


Figure 1. Sub-detectors within the sPHENIX detector (sPHENIX Collaboration, 2022)

measure the energy of hadrons, making the calorimeters vital to sPHENIX's jet physics goals (Figure 2). The EMCAL is comprised of 24,576 tungsten-epoxy towers, with 567 Scintillating fibers embedded in each tower. The EMCAL is read out by silicon photomultipliers whose analog output signal is wave-form digitized into 12 individual time samples. These waveforms are then fitted with a template to extract the energy, the time, and the pedestal of each calorimeter hit. [4]

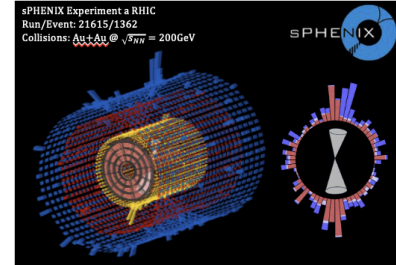


Figure 2. Di-jet observation in the sPHENIX Calorimeters (Ma, 2024)

iii. Waveform fitting

As mentioned above, extracting pedestal, time, and energy values from EMCAL waveforms requires fitting each pulse with a template (Figure 3). Although the fit can accurately extract these values, the large number of EMCAL towers and the unprecedented number of events means that significant computing resources are required to process the data. [4]

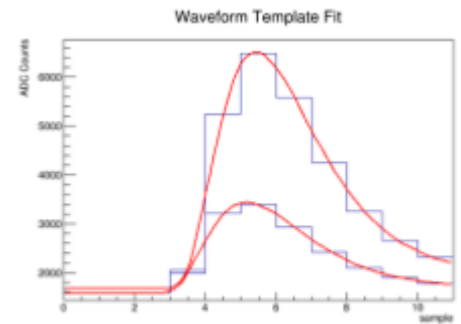


Figure 3. Waveform Histogram with template fit

Picture taken by Om Patel and Zen Pinkenburg using ROOT

iv. Machine Learning

Since the waveform fitting has limitations in speed and accuracy at low energy pulses, we investigated the use of a machine learning algorithm to analyze the EMCAL waveforms [3]. Machine learning algorithms excel at recognizing patterns in data, so we aim to provide a better resolution for waveforms containing noisy data and faster reconstruction speeds with the implementation of our machine learning algorithm. The neural network trains with data from several million fitted waveforms and receives the twelve-time samples as inputs and outputs each waveform's time, energy, and pedestal values.

Methods

i. Filtering Data

To investigate the feasibility of applying machine learning the waveforms and the fit results were used to train a machine learning model. The raw waveforms contain noisy and empty channels (Figure 4a). To simplify the training, cuts to the data were applied to ensure that only high-quality waveforms were used to train the model. The cuts applied include a cut on the minimum to get rid of empty channels, a cut on the energy (maximum – pedestal) to remove very small signals, and a cut on the first, last, and middle bins to remove noisy channels. This resulted in much cleaner waveforms which were then used to train the model (Figure 4b). The cuts

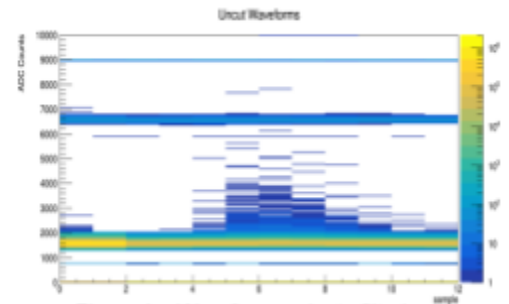


Figure 4a. Waveforms prior to filtering

Picture taken by Om Patel and Zen Pinkenburg using ROOT

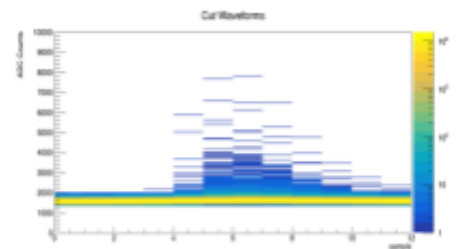


Figure 4b. Waveforms after cuts on amplitude, minimum, and pedestal. This cleaned data was used to train the model

Picture taken by Om Patel and Zen Pinkenburg using ROOT

mainly removed empty channels and resulted in a ~98% reduction in the number of waveforms and we therefore used 10M events to train the model.

ii. Machine Learning Architecture

The model consists of an input layer, a normalization layer, five dense layers with ReLU activation, and a final dense layer with a linear activation function (Figure 5). The input layer receives the raw data, which consists of 12 features (inputs). The normalization layer standardizes the input data to help the model train more effectively. In the dense layers, each neuron takes inputs from the previous layer, applies weights and a bias, and passes the result through the ReLU activation function. ReLU introduces non-linearity, allowing the network to model more complex patterns. The output of each neuron in one dense layer is then fed forward as input to the neurons in the next layer. The neurons' collective behavior allows the network to learn patterns in the data and make accurate predictions. The final dense layer uses a linear activation function to combine the processed information and returns 3 predicted values. The

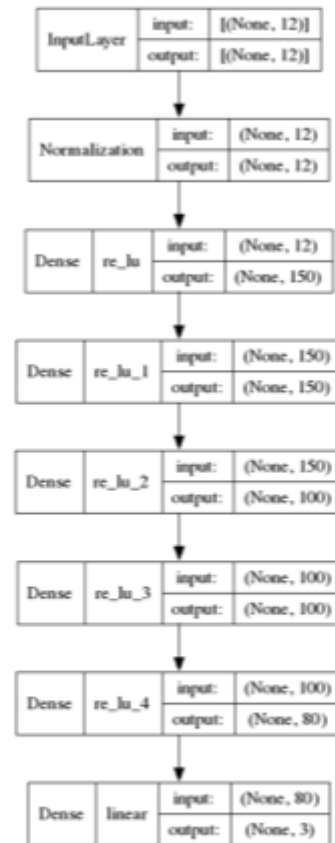


Figure 5. Architecture of machine learning model

Image created by Om Patel and Zen Pinkenburg using Canva

optimizer used was Adam and the loss function was mean squared error. The machine learning algorithm takes approximately 0.339 ms to predict each fit.

iii. Implementation in Reconstruction

Since the model is written in Python and the data is reconstructed by a C++-based framework, a temporary solution was implemented to execute Python code from within the reconstruction. For each event, the waveform values were written to a file, which was then given to the model for predictions. The model returned the predicted energy, time, and pedestal values via another file. To gather sufficient statistics, this was run on the sPHENIX 60,000-core condor batch farm.

Results

i. Pedestal, Time, and Energy Predictions

To compare the results, six histograms were generated using ROOT to display the fit-calculated and predicted time, pedestal, and energy values (Figure 6). The histograms have similar shapes, means, and standard deviations, which shows that the model's predictions are very close to the fit's values. However, this only confirms the model's precision, not its accuracy.

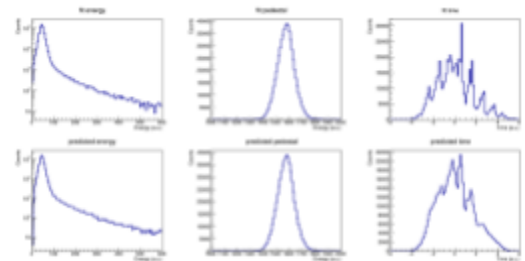


Figure 6: (From left to right) fitted energy, pedestal, and time (top), and predicted energy, pedestal, and time (bottom)

Picture taken by Om Patel and Zen Pinkenburg using ROOT

ii. Energy Comparison

When the predicted and fitted energy values are directly compared to each other, they show a strong linear correlation (Figure 7). The statistical errors are smaller than the symbol size. The resulting line of best fit (shown in red) has a slope of 1.04219, which signifies that the model is very precise in the energy predictions (a slope of 1 would mean that the model predicts the energy perfectly). At larger energies, however, the model deviates more from the fitted energy, as shown by the residuals. This is because the model is inaccurate at predicting values outside of its training set, so predicting rare higher energy values is less accurate.

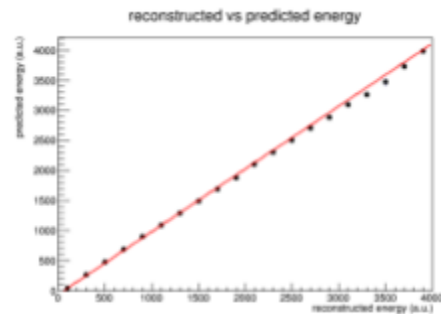


Figure 7: fitted energy (x-axis) and predicted energy (y-axis) with linear regression line in red

Picture taken by Om Patel and Zen Pinkenburg using ROOT

iii. π^0 Reconstruction

A π^0 is a particle that is commonly produced in heavy ion collisions at relativistic energies. When a π^0 decays, it produces two back-to-back photons, whose energies are then measured by the EMCAL. If the two photons hit separate EMCAL towers, the energy produced by the waveform fit of each tower can be used to reconstruct the π^0 s, with the width of the π^0 peak governed by the energy resolution of the EMCAL.

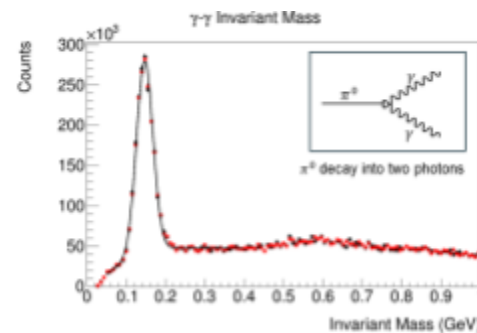


Figure 8. Red represents the invariant mass reconstructed from fitted waveforms and black from waveforms as predicted by the model

Picture taken by Om Patel and Zen Pinkenburg using ROOT

Thus, by using the energy predicted by the model, the π^0 reconstruction provides a direct

method to evaluate the model's accuracy. As seen in Figure 8, the invariant mass distribution is almost identical when using the template fit or the model to extract the tower energies.

Discussion

sPHENIX's EMCAL is vital to achieving the collaboration's jet physics goals. Currently, a template-fit method is used to extract the pedestal, time, and energy values from the EMCAL's digitized waveforms. However, this approach is limited in speed and accuracy with small pulses. To improve on the reconstruction process, we investigated the use of a machine learning model to predict the pedestal, time, and energy values. The results of the machine learning were very promising, showing a high degree of accuracy in its predictions, but more work is required before it can be fully integrated into sPHENIX's reconstruction software.

Conclusion

The model was able to make accurate predictions of the energy. However the model's prediction was slower than the waveform fit (averaging over the same 100 events, the machine learning took ~ 255 ms per event, whereas the fit took ~ 140 ms). However, the machine learning algorithm was not optimized for speed, so it should be possible to decrease the running time significantly by optimizing the inference process. In addition, the implementation of the model into the reconstruction currently requires a restart for every event which takes a large amount of time. Moving to the ONNX software package should allow us to use a C++ API which can then be integrated into the sPHENIX software stack.

Future Research

i. ONNX

ONNX is a software package that allows developers to use models and frameworks across different programming languages. Our current method of incorporating the machine learning algorithm within the C++-based sPHENIX reconstruction framework is inefficient. ONNX would resolve this issue by removing the need to open and close the input file in the macros for each event. During the project development, we attempted to use ONNX to convert our model to use it directly in C++. We tried to use several ONNX versions, but each ONNX version has different dependencies and package version requirements, which made configuring each version time-consuming and tedious. We plan a systematic effort to create a working ONNX implementation that can be run on the sPHENIX batch farm.

ii. Hadronic Calorimeters

Since the EMCAL and the hadronic calorimeters use identical readout electronics, analyzing digitized waveforms from the hadronic calorimeters would be easy, as the procedure would only vary slightly. We would achieve this by retraining the machine learning model to predict hadronic signals using the template fit results for the hadronic calorimeters.

iii. Pushing the Model

As of now, our model predicts high-quality waveforms as shown in Figure 4b. So, we plan to test the model in situations where waveform fitting becomes challenging, such as with small

energies, non-centered pulses, or unusual patterns caused by readout electronics failures.

Ultimately, we aim to have the model predict these non-ideal waveforms and cover ranges where the template fitting approach reaches its limits.

Acknowledgments

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